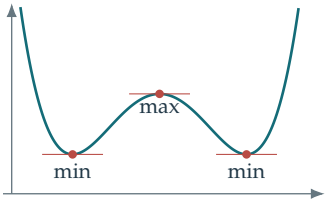
Panel 1

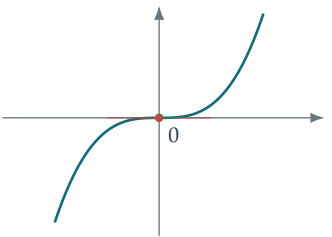
Local extrema



$f'(x) = 0$ at each marked point

Panel 2

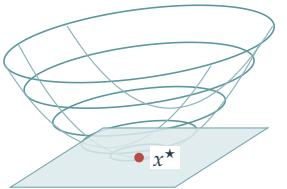
Stationary inflection



$f'(0) = 0$: neither min nor max

Panel 3

Convex objective



$\nabla f(x^\star) = 0$

A global minimizer